

High-Level *Design*

Semantic Search, Recommendation System & AI-Based Business Analysis — system context, architecture, data flow, and non-functional requirements.

DOCUMENT HLD v1.0	SCOPE Whole Platform	AUDIENCE Architects · Leads
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01

Objective & Scope

WHY

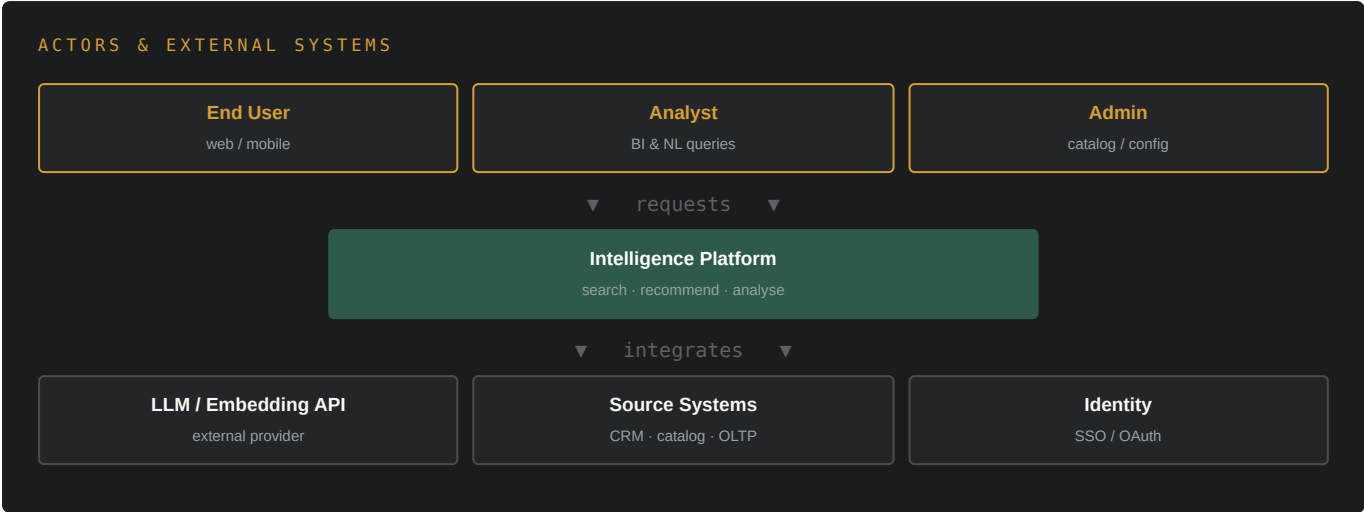
The platform delivers three intelligence capabilities over a shared data and embedding foundation. The goal is to reuse infrastructure — a single set of embeddings and one event log serve all three services rather than each maintaining its own stack.

- **Semantic Search** — retrieve documents/products by meaning, not keywords.
- **Recommendations** — personalised item suggestions from behaviour and content.
- **AI Business Analysis** — dashboards, forecasting, and a natural-language analytics assistant.

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System Context

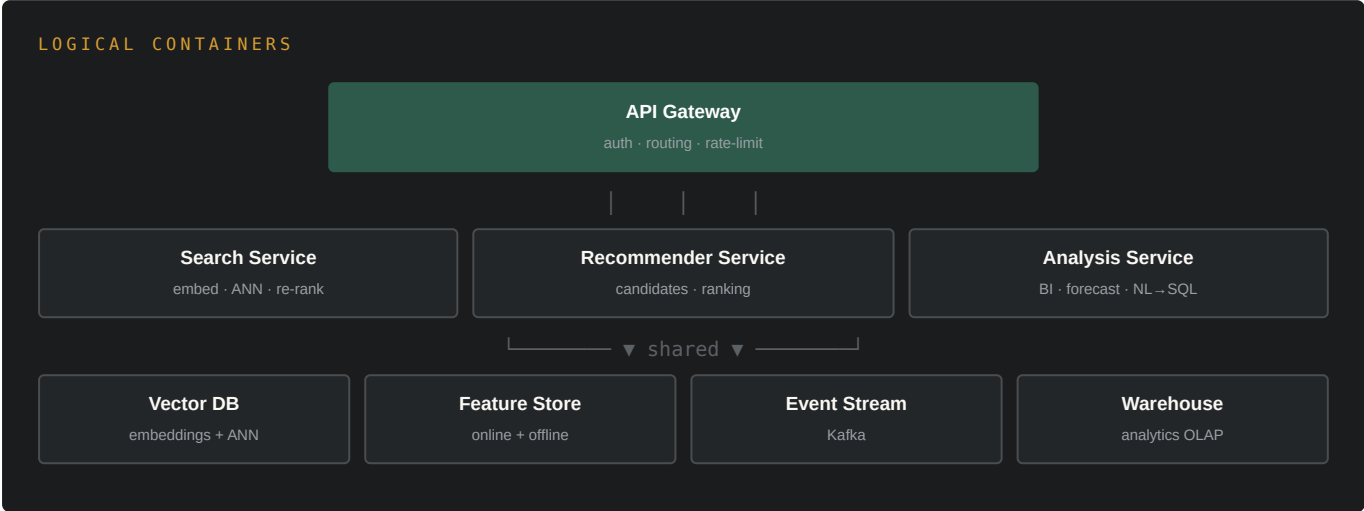
C4 · LEVEL 1



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Container Architecture

C4 · LEVEL 2



Design principle — embeddings are computed once at ingest and shared. The Search Service owns embedding generation; the Recommender reuses those vectors for content-based candidates; the Analysis assistant reuses them for retrieval-augmented answers.

Ingest (write path)

Source systems → CDC/ETL → clean & chunk → embedding model → vectors land in Vector DB; structured rows land in Warehouse; item features land in Feature Store.

Query (read paths)

- **Search** — query embedded → hybrid ANN + BM25 → cross-encoder re-rank → results.
- **Recommend** — user context → candidate generation → feature enrichment → ranking model → ordered list.
- **Analyse** — NL question → LLM plans SQL / RAG → warehouse query → summarised answer + chart.

Feedback loop

All user interactions (clicks, views, purchases, query reformulations) flow back through the Event Stream into the Feature Store and Warehouse, continuously feeding model retraining.

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Component Responsibilities

OWNERSHIP

COMPONENT	RESPONSIBILITY	KEY TECH
API Gateway	Authentication, routing, throttling, request shaping	Kong / FastAPI edge
Search Service	Embedding generation, hybrid retrieval, re-ranking	FastAPI, cross-encoder
Recommender Service	Candidate gen, feature lookup, ranking, fallback	FastAPI, LightGBM/NN
Analysis Service	BI APIs, forecasting jobs, NL → SQL agent, RAG	LLM, dbt, Prophet
Vector DB	Store/query embeddings via ANN index	Qdrant / pgvector
Feature Store	Serve online features <10ms; offline for training	Feast + Redis
Event Stream	Durable interaction log, fan-out to consumers	Kafka
Warehouse	Analytics store, source of truth for BI	BigQuery / Snowflake
Orchestrator	Schedule ETL, retraining, batch embedding	Airflow

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Non-Functional Requirements

QUALITY

PERFORMANCE Search p95 < 300ms Recommendations p95 < 200ms; analysis NL answers < 8s.	SECURITY SSO, RBAC, row-level PII masking in warehouse; secrets vaulted.
SCALABILITY Horizontal, stateless services Vector DB & Kafka partitioned; autoscale on QPS.	OBSERVABILITY Traces · metrics · logs OpenTelemetry; model quality dashboards.
AVAILABILITY 99.9% read path Graceful degradation to popularity / keyword fallback.	COST Cache & batch embeddings Reuse vectors; cache hot queries & LLM responses.

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Key Risks & Mitigations

WATCH

RISK	MITIGATION
Cold start for new users/items	Content embeddings + popularity fallback until behaviour accrues
Embedding model drift / version skew	Version vectors; re-embed on model change; shadow eval
LLM hallucination in NL → SQL	Constrain to validated schema; show generated SQL; guardrails
Vector DB scaling limits	Sharding + ANN tuning; offload cold vectors